

PUSHKARAJ BARADKAR

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EDUCATION

University of Southern California, Viterbi School of Engineering GPA: 3.73 / 4.00

Jan 2025 - Dec 2026

Master of Science in Computer Science

Los Angeles, CA

Machine Learning for Data Science, Analysis of Algorithms, Web Technologies, Database Systems

University of Mumbai GPA: 9.13 / 10

Jul 2020 - May 2024

Bachelor of Engineering in Computer Engineering

Mumbai, India

Data Structures, Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing

EXPERIENCE

AI Engineer Intern | Tabhi, Austin, TX

May 2026 – Present

- Built a content-based recommendation prototype: multi-modal embeddings with FAISS cosine retrieval over a 188K-item catalog, plus a recency-decayed, event-weighted user-vector layer built as a drop-in slot for later two-tower/SASRec models.
- Designed a retrieval, rank & re-rank funnel (ANN retrieval, weighted-sum ranking) on a pgvector for 350K users and 500K items.
- Built a leave-last-out eval harness (Recall@K, MRR@K, coverage) with popularity and co-occurrence baselines.
- Evaluated retrieval with Recall@100 and NDCG@10, targeting relative lift of two-tower ranking over the content baseline.
- Optimized the pipeline for memory-constrained hardware via chunked batch encoding, preallocated buffers, and explicit cleanup.

Software Engineer Intern – Real-Time AI Systems | USC, Los Angeles, CA

Nov 2025 – May 2026

- Built a real-time SLAM pipeline combining monocular VO and depth estimation, achieving ~20–25 FPS on noisy indoor data.
- Implemented RANSAC-based feature matching (2500 matches, 2000 inliers), leveraging IMU to improve motion consistency under noise.
- Resolved scale ambiguity using sparse depth and fused VO with GPS in GTSAM, reducing trajectory error from ~95 m to ~1.2 m ATE.
- Debugged sensor noise, calibration drift, and VO failures, improving tracking stability and reducing dropouts by ~30%.

Software Engineer Intern – Backend & APIs | Technoriya ERP Solution, India

Oct 2023 – Dec 2023

- Developed and optimized backend services and APIs using Python and Node.js, supporting scalable data pipelines, system reliability, cloud-based deployment and processing layers for downstream analytics and ML workflows.
- Implemented secure authentication and access protocols via Firebase Auth and Firestore rules to ensure data integrity.
- Optimized RESTful APIs and Firestore queries to achieve 450 ms response times under 1,000 concurrent requests for ML integration.
- Streamlined deployment workflows using Vercel and GitHub Actions to improve delivery consistency and reliability.

PROJECTS

Clinical LLM: Domain-Adapted Medical QA with Optimized Inference | PyTorch, Hugging Face, vLLM

May 2026 – Present

- Fine-tuned Llama 3.2 3B for clinical Q&A using LoRA with 4-bit quantization on de-identified datasets (MedQA, PubMedQA), raising MedQA accuracy from 41% to 59% on a single 24GB GPU.
- Cut p95 latency 2.3x and lifted throughput 3.1x via vLLM; paged KV caching, FlashAttention-2, speculative decoding with a 1B draft model.
- Built eval/serving pipeline benchmarking accuracy, latency, memory, exposed behind a FastAPI/vLLM endpoint with continuous batching.

Godot MCP: AI-Native Game Development Interface | TypeScript, Node.js, MCP - [Link](#)

April 2026 – Present

- Built an MCP server enabling LLM agents (Claude, Cursor) to control Godot projects via natural language; used by 10+ active users.
- Developed 30+ tools for scene, script, execution, and asset management with secure path validation and structured routing.
- Implemented CI/CD, auto-config detection, and multi-client integration for end-to-end 2D/3D workflow automation.

AgentPay: Autonomous Agent Economy | Solana, Avalanche, LangChain, Next.js - [Link](#)

April 2026 - Present

- Built a multi-agent system where AI agents autonomously discover, hire, and pay other agents using on-chain payments and reputation.
- Designed smart contract system (Rust/Anchor) for agent registry, escrow, and staking/slashing with verifiable reputation tracking.
- Integrated LangChain-based orchestration with custom payment tooling (x402), enabling under 30s end-to-end agent interaction cycles.

Open Source Contribution – Defillama SDK (RPC Reliability & Failover) | TypeScript, Node.js, Web3 - [Link](#)

April 2026

- Fixed RPC override bug (Fixes #162) by redesigning Chain RPC logic to correctly prioritize user-defined endpoints.
- Implemented runtime RPC quarantine (5-failure threshold, 10 min cooldown), reducing repeated retries and improving request reliability.
- Added 8 unit tests covering override and failure-handling logic; expanded SDK documentation for RPC configuration.

TECHNICAL SKILLS, PUBLICATIONS, CONFERENCES

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript
- **AI / ML & Data Science:** Deep Learning, Generative AI, NLP, LLMs (ChatGPT, Claude Code, Copilot), Transformers, RAG, Agentic AI Systems, Autonomous Agents, Prompt Engineering, Computer Vision, Data Mining, Data Science, PyTorch, TensorFlow, LangChain, Pandas, NumPy, Matplotlib, LangGraph, MCP, Google ADK (Agent Development Kit), A2A
- **Frameworks & Development:** FastAPI, Flask, Django, Node.js, Express.js, React.js, Next.js, Angular.js, OAuth, OOP, Agile, HTML/CSS
- **Databases & Cloud:** MongoDB, SQL, PostgreSQL, Supabase, Redis, AWS, GCP, Azure, Docker, Kubernetes, Git, CI/CD, Gitlab, Linux
- **Software Engineering:** Object-Oriented Programming (OOP), REST APIs, debugging, Agile, REST API, AWS, Lambda, EC2